Employee Onboarding Checklist Tracker

Backend Development Assignment - Case Study 183

B.Tech Computer Science Engineering

Node.js, Express.js and MongoDB

1. Introduction

The Employee Onboarding Checklist Tracker is a web application that helps an HR team manage tasks for new employees. Common tasks include collecting an ID card, receiving a laptop, completing an employee profile, and attending orientation. The application keeps these tasks in one place and shows how much of the onboarding process has been completed.

The project uses a simple Express backend and a React frontend. HR administrators assign tasks, while employees can sign in and update the completion status of their own tasks.

2. Objectives

- Create Employee and OnboardingTask schemas using Mongoose.
- Provide create, read, update, and delete operations for checklist tasks.
- Allow only HR administrators to create employee accounts and assign tasks.
- Allow task completion only by the employee concerned or the assigning HR admin.
- Validate input and calculate onboarding progress in the backend.

3. Technologies Used

Technology	Use in the project
Node.js and Express.js	Run the backend and provide REST API routes.
MongoDB and Mongoose	Store employee and task records with schema validation.
JWT and bcryptjs	Authenticate users and hash account passwords.
React and Vite	Build and run the user interface.
Postman	Send API requests and check responses.

4. Basic Working

The user logs in with an email address and password. After successful login, the backend returns a JWT. The frontend sends this token with protected requests. HR can create an employee account, open its checklist, and add tasks with due dates. The employee marks tasks complete, and the backend recalculates progress from the saved records.

Application flow: React interface -> Express API -> Mongoose -> MongoDB.

5. Database Design

The database uses two collections. Each task references both the employee who must complete it and the HR administrator who assigned it. One employee can have many onboarding tasks.

Employee collection

Field	Description
name	Employee or HR administrator name.
email	Unique email address used for login.
password	Hashed password; not included in normal API responses.
department	Department of the account holder.
role	Either hr or employee. New employee accounts use employee.

OnboardingTask collection

Field	Description
employee	Reference to the employee receiving the task.
assignedBy	Reference to the HR administrator who created the task.
taskName	Description of the task, up to 120 characters.
status	pending or completed; the default is pending.
dueDate	The date by which the task should be completed.

Both schemas also store creation and update timestamps.

6. Main API Endpoints

The following routes use the base path **/api**. Protected routes require a Bearer token.

Method	Endpoint	Purpose
POST	/auth/login	Sign in and receive a JWT.
GET	/auth/me	Get the current account.
GET / POST	/employees	List employees or create an account (HR).
GET / POST	/employees/:id/checklist	Read a checklist or assign tasks.
GET	/employees/:id/onboarding-progress	Get total, completed, pending, and percentage.
PATCH	/tasks/:id/complete	Complete or reopen a task.
PATCH	/tasks/:id	Edit a task name or due date.
DELETE	/tasks/:id	Delete a task.

7. Authorization and Validation

JWT authentication is checked before protected routes run. HR can view employee checklists and progress across the team. An employee can only view their own checklist. Only the HR administrator who assigned a task can edit or delete it. That HR administrator or the assigned employee can change its completion status.

The API checks required account fields, email format, password length, task names, valid dates, and completion values. Duplicate email addresses are rejected. Invalid input returns a suitable error response, while unauthorized requests are rejected with status 401 or 403. Passwords are hashed before saving.

8. Frontend Screens and Progress

- **Login:** accepts an email address and password and displays login errors.
- **HR Dashboard:** lists employees, their departments, and onboarding progress. It also includes employee creation and search.
- **Employee Progress:** lets HR view a checklist and manage tasks they assigned.
- **My Checklist:** lets employees view their tasks and toggle completion.

Progress is calculated as completed tasks divided by total tasks, multiplied by 100. The result is rounded down to a whole number. For example, 2 completed tasks out of 4 gives 50%. An empty checklist shows 0%, and 100% is shown only when every task is complete.

9. Testing and Results

The following checks were completed during project development:

Check	Observed result
Login and session validation	Valid login succeeded; invalid or expired credentials were rejected.
Checklist operations	HR could add, view, edit, and delete assigned tasks.
Ownership restrictions	Another employee or unassigned HR could not change the task.
Input validation	Invalid names, dates, IDs, and empty checklists were rejected.
Saved progress	Completion changed progress and remained saved after reload.
Frontend and API checks	Production build passed. Desktop/mobile screens and Postman requests were checked.

10. Conclusion

The project meets the main requirement of tracking onboarding tasks with role-based and ownership-based access. It gives HR a clear view of employee progress and gives employees a simple checklist to follow. The backend remains easy to understand by using separate models, middleware, and route files. Email reminders and password reset could be added in a future version.

Reference: Node_Final.pdf - Employee Onboarding Checklist Tracker, Case Study 183.